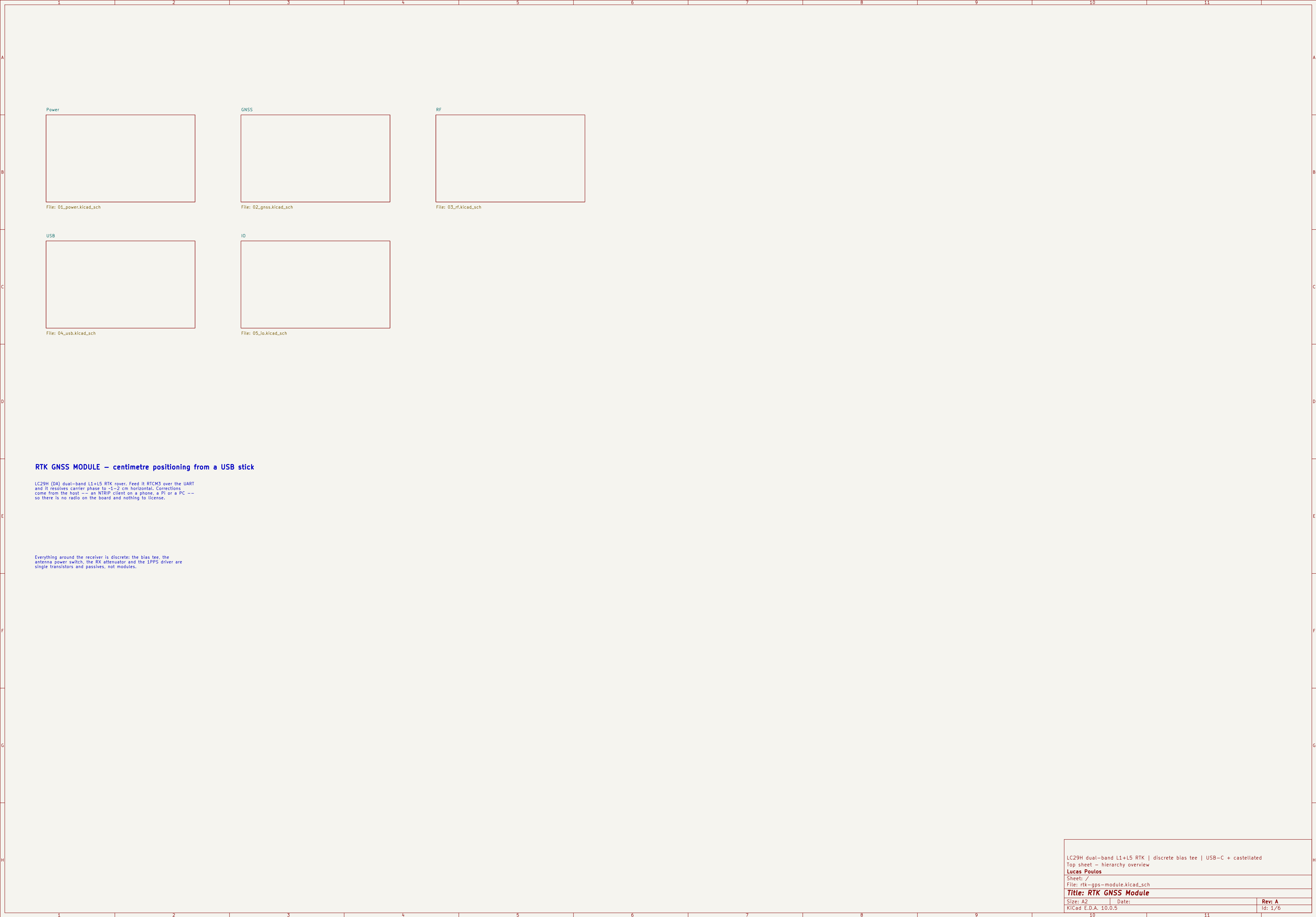
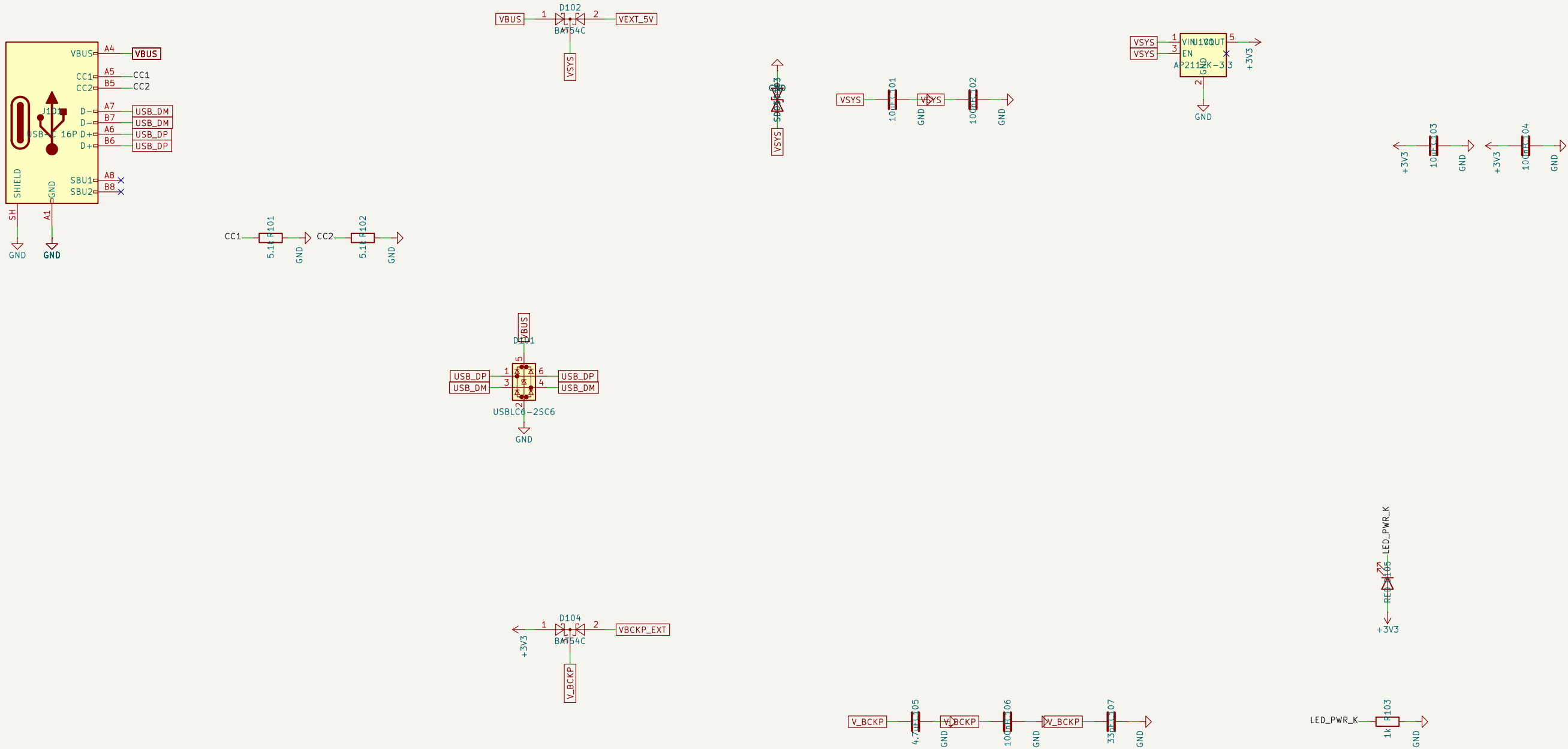




SHEET 1 – INPUT, 5 V ORING, 3V3 AND THE BACKUP DOMAIN

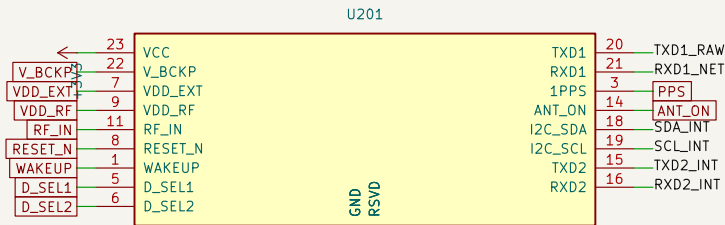
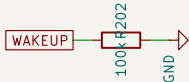
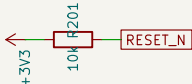
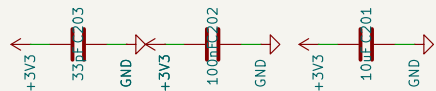


D102 ORs USB VBUS against the carrier board's 5 V so the two can be present at once without either back-feeding the other. One BAT54C is two Schottkys; a shared cathode is exactly the OR. Cost of the diode drop: VSYS = 4.7 V, and the LDO needs only 3.62 V to hold 3.3 V at 600 mA.

D104 does the same job for the backup rail. V_BCKP holds the RTC and the ephemeris cache; keep it alive and the receiver hot-starts in ~1 s instead of cold-starting in ~30 s. With nothing on VBCKP_EXT it simply follows 3V3. C105/C106/C107 are the 4.7uF + 100nF + 33pF the LC29H hardware design asks for on this pin (section 3.2).

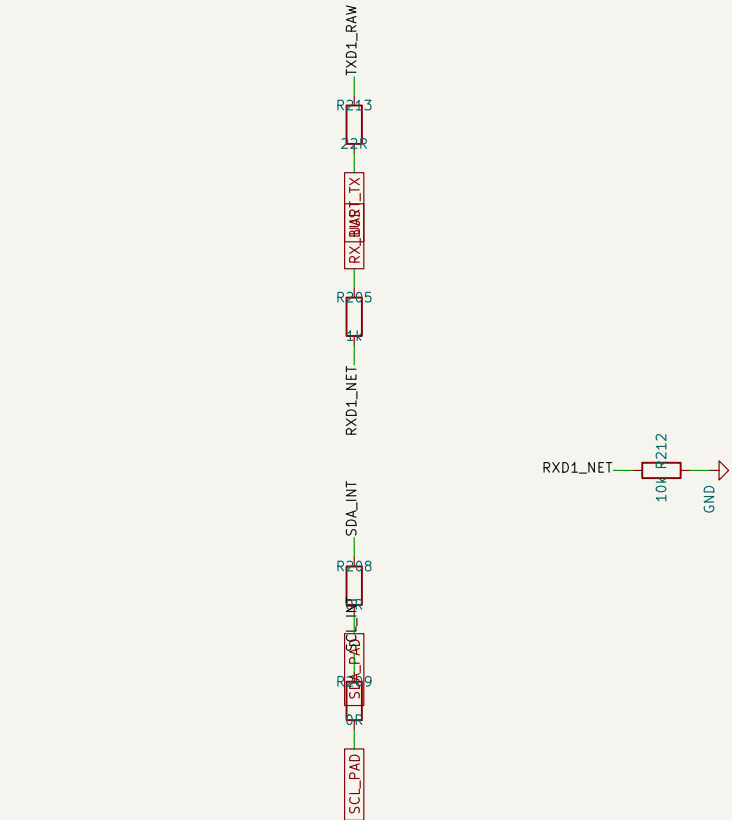
Wired		
Sheet 2: Power		
Lucas Poulos		
Sheet: /Power/		
File: 01_power.kicad_sch		
Title: USB input, 5 V ORing, 3V3 LDO, backup domain		
Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 2/6

SHEET 2 – LC29H (DA) DUAL-BAND RTK RECEIVER



D_SEL1/D_SEL2 pick the host interface. Both are pulled down internally by 75k, and 0.0 selects UART1 -- which is what this board wants, so the default needs no parts. R206/R207 are DNP links to a 1.9 V node for the other three combinations. The 1.9 V comes from dividing VDD_EXT 2.8 V by 4.7k/10k: tying D_SEL to VDD_EXT directly would exceed VIHmax = 2.1 V.

RX LEVEL. RXD1 is specified VIHmax = 3.08 V (Table 6) and the CH340N drives a full 3.3 V. R205/R212 divide by 10/11 -> 3.0 V, inside spec and still far above the 1.75 V VIHmin. The divider sits AT the module pin, so it protects the castellated RXD1 pad and the USB bridge with one pair of resistors instead of one pair per source. Source impedance 909 R into -15 pF is a 14 ns edge -- negligible against a 1.09 us bit at 921600 baud.

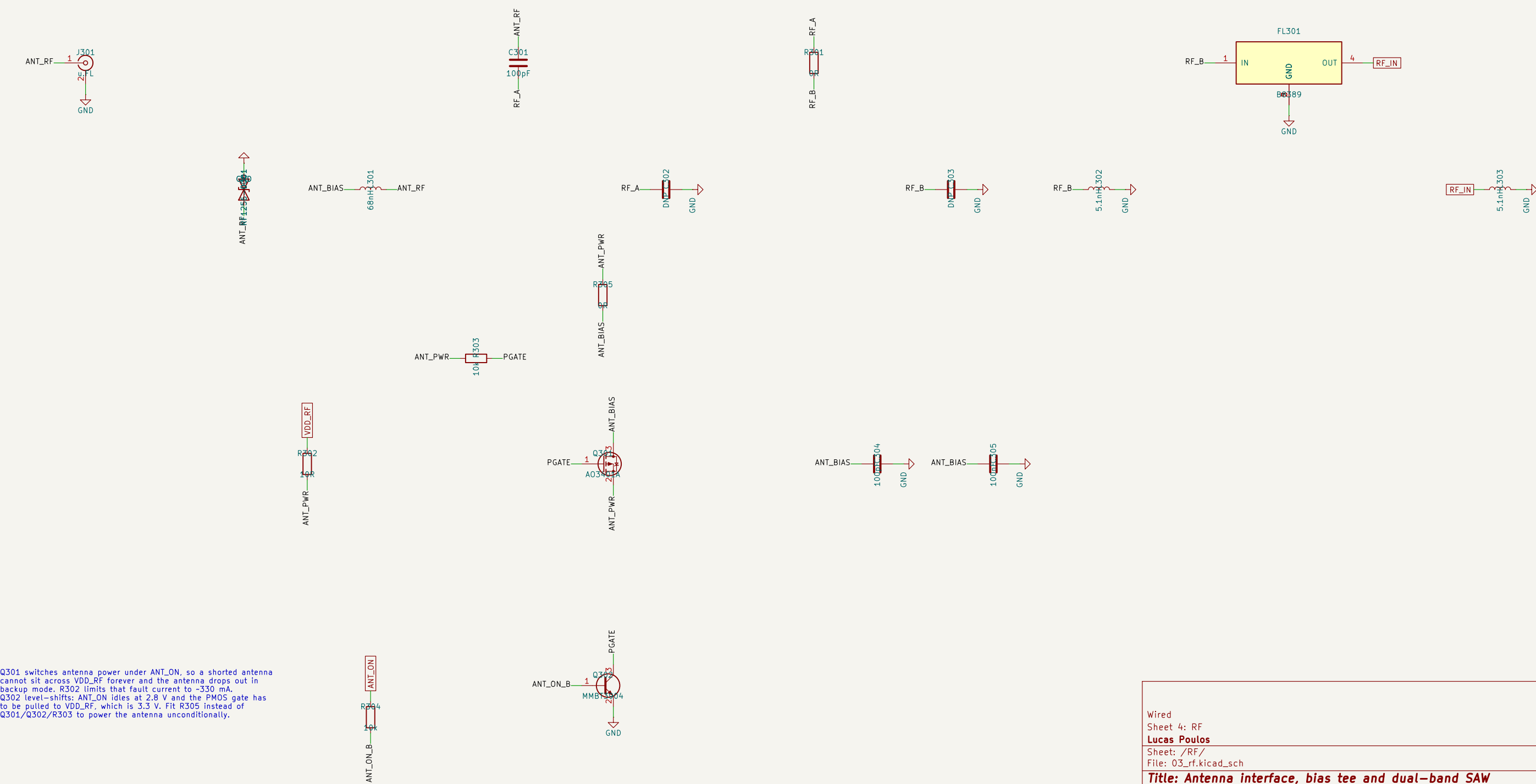


Wired	
Sheet 3: GNSS	
Lucas Poulos	
Sheet: /GNSS/	
File: 02_gnss.kicad_sch	
Title: LC29H (DA) receiver, decoupling and strapping	
Size: A3	Date:
KiCad E.D.A. 10.0.5	Id: 3/6

SHEET 3 – 50 OHM ANTENNA FRONT END AND BIAS TEE

The whole chain from J301 to RF_IN is a 50 ohm line. L301 injects DC onto it without loading it; 68nH is 673 ohm at L1 and 502 ohm at L5, both high against 50 ohm. C301 blocks that DC out of the module. R301/C302/C303 are a pi network left as 0R + two unfitted pads -- somewhere to retune if a particular antenna misbehaves, at no cost if it does not.

FL301 is the dual-band SAW Quectel names in its own reference design. It passes 1176.45 MHz (L5/E5a/B2a) and 1559-1607 MHz (L1/E1/B1/G1) and rejects 20-39 dB everywhere else. An L1-ONLY SAW here would be a bug: it would delete the second frequency that makes this an RTK receiver rather than a plain GPS. L302/L303 are the 5.1 nH shunts the SAW needs to actually present 50 ohm -- its ports are 50 ohm // 5.1 nH, not 50 ohm.



Q301 switches antenna power under ANT_ON, so a shorted antenna cannot sit across VDD_RF forever and the antenna drops out in backup mode. R302 limits that fault current to ~330 mA. Q302 level-shifts: ANT_ON idles at 2.8 V and the PMOS gate has to be pulled to VDD_RF, which is 3.3 V. Fit R305 instead of Q301/Q302/R303 to power the antenna unconditionally.

Wired		
Sheet 4: RF		
Lucas Poulos		
Sheet: /RF/		
File: 03_rf.kicad_sch		
Title: Antenna interface, bias tee and dual-band SAW		
Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 4/6

